PhaseAtlas

SCIENTIFIC SOFTWARE
Niu Te | Imperial College London

Interactive PDE research platform

Editable equations / 3D space-time surfaces / numerical verification

Research problem

Researchers repeatedly build separate tools to simulate a PDE, visualize its evolution and check numerical accuracy. PhaseAtlas brings these tasks into a browser workspace, with editable equations, explicit boundary data and saved trajectories. It supports model exploration, numerical-method development and teaching through inspectable, reusable experiments.

Workflow and contribution

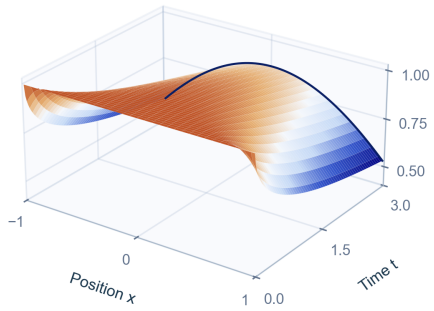
Define a model, run it locally, inspect the solution and compare resolutions. The platform links synchronized animation and 3D surfaces to parameter scans, boundary-balance checks and travelling-wave diagnostics. Exported records retain equations, parameters, numerical settings and arrays so collaborators can replay and extend a calculation.

1D / 2D
intervals and rectangles

3D x-t-u
rotation, playback and PNG export

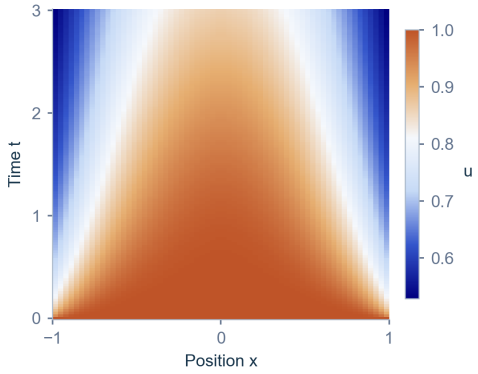
96 tests passed
mathematical and numerical checks

A | 3D space-time surface



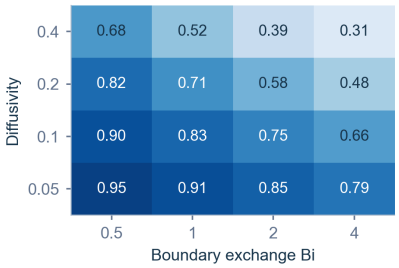
Height and colour encode u(x,t); saved frame highlighted.

B | Space-time heatmap

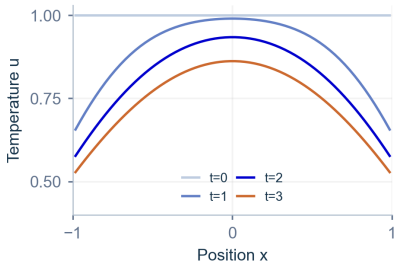


The same field on a fixed colour scale.

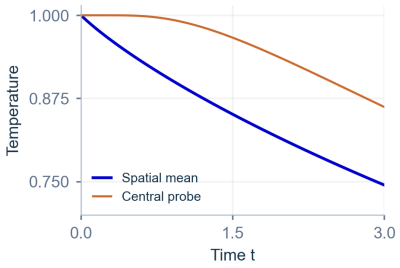
C | Final mean at t = 3



D | Solution profiles

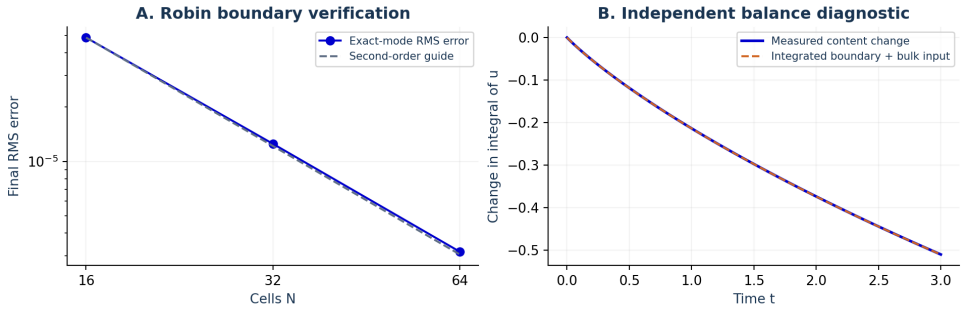


E | Mean and local probe



A, B, D, E: diffusivity 0.1; Bi = 2; N = 64; time step 0.001. C: 16 runs on N = 32. Dimensionless units.

F-G | Exact-mode refinement and boundary-balance verification



Observed order

p = 1.99995

Robin eigenmode: N = 16, 32, 64; fixed time step 0.0002. Final RMS error falls from 4.86e-5 to 3.15e-6.

Methods, applications and reuse

Models. One to three coupled fields; periodic Fourier/ETDRK4 models in 1D/2D; bounded diffusion with Dirichlet, Neumann and passive Robin conditions; 1D fourth-order equations with two configurable linear boundary rows per end. **Checks.** Time/grid refinement, supplied-reference residuals, conservation and boundary diagnostics. **Reuse.** JSON/CSV trajectories, SVG/PNG figures and WebM animations. The illustrated dimensionless cooling model compares internal diffusion and surface heat exchange; numerical computation runs in Float64 browser workers.

DEMO phaseatlas.onrender.com CODE github.com/NIU-55383/PhaseAtlas